Soup Training Monitor

Frontend delivery — 10 November 2026. Production · Repository. Five screens: Dashboard, Training runs, Run details, Compare and Data quality. All data is labelled demonstration snapshots. The separate Soup landing deployment is unchanged.

Product and truthful charts

Twenty deterministic scenarios include NaN/Infinity, duplicate conflicts, unsorted steps, cadence gaps, zero, tiny values and a singleton. R05 has 1,000,000 rows and one unique Loss maximum: sourceIndex 500001, step 500001, value 100. Four raw Float64 columns remain Worker-owned. Independent per-metric reduction preserves exact source coordinates and visible finite extrema. Conflicts and known gaps break the authoritative line; source members remain inspectable. No smoothing or interpolation is represented as a measurement.

Dashboard includes a focused monitor and interactive run/metric map with text alternatives. Charts support wheel/pinch zoom, Shift-wheel pan, drag and keyboard range controls. Details and Compare share one stable source table with all three raw metrics, actual steps, distance and source-qualified Pin. Whole-run facts remain independent of the viewport; NaN never becomes zero. No scientific training-stop recommendation is fabricated.

Verification and evidence

Source commit 41ae67f48be6364f719ba38ad71d694c8ae484fe; source/build hashes are in reports/build-metadata.json. Build, lint, 106 unit tests (27 files), 113 Chromium acceptance tests and regeneration of all 20 manifest entries pass. No skipped/flaky browser cases or retries. Raw oracles, controlled stale-response races, locale/timezone, keyboard, synthesized touch, responsive layouts, DPR2, actual Canvas spike/gap pixels and cleanup are covered. Six injected fault classes fail named assertions; failed results remain archived.

Thirty axe audits report zero violations. All 528 fresh contrast incomplete occurrences have individual screenshot/pixel dispositions (minimum 5.096:1); the original 516 are also retained. Actual Canvas/SVG contrast is checked in both themes. Separate Lighthouse Performance is 94–98, Accessibility/Best Practices 100. This is assistant-run evidence, not human screen-reader certification. Tests cannot prove every oscillation, every browser or scientific logger validity.

Performance and verdict

Windows 11, Ryzen 7 8845HS, Chromium158, 1440×900, DPR1. Main renderer CPU×4; Worker throttling unsupported. Each run uses five cold contexts and 30 detail plus 30 four-run Compare range/hover pairs, nearest-rank p95, no forced GC. Two same-source full Chrome recordings pass all expanded targets. Worst p95: cold 2531.9ms; detail/Compare zoom 114.8/131.4ms; detail/Compare hover 40.6/48.6ms. Matched previous-source full-trace baseline: 2493.6ms cold, 149.6/214.7ms zoom, 88.5/107.5ms hover. Cold confirmation is slightly slower, within budget. Separate no-trace results are not pooled.

Hover retains the original browser pointer timestamp through current source DOM and two RAF callbacks: a rendering opportunity, not physical monitor presentation. Exact bounded source caches, unchanged-row suppression, shared readouts and offscreen overlay deferral reduce work. Native crosshair/drag remains. Traces, raw samples, failed experiments and before/after source identities are retained. Shared-machine sequential measurements limit causal inference.

SHIP for the implemented demonstration frontend, conditional on the generated evidence manifest having no pending checks and exact published-byte verification. The original PDF has no numeric ceilings; the expanded latency targets now pass. Longer-session retained-memory analysis is the first next improvement, not an unverified claim of leak freedom.

What surprised me and still concerns me

Raw lookup was already fast; duplicated DOM updates and frame work dominated hover. A genuine MutationObserver test caught repeated unchanged-row writes, and report HTML unexpectedly changed Tailwind output until source scanning was bounded. Native renderer/Worker footprint at 1M is 145.9MiB private committed /245.6MiB working set, growing to 319.5/588.7MiB after 20 transitions. Raw columns are 32,000,000 bytes; indexed owned buffers 76,696,612 bytes including raw. These scopes are not additive. Shared GPU/browser memory is separate. Settled lifecycle checks observe one Worker/three charts and zero Workers after close; memory growth remains a concern requiring retained-object analysis.

AI assistance and handoff

AI assisted architecture, code, independent oracles, fault injection, browser automation, primary-source research and profiling. Suggestions were accepted or revised against original identities, actual Canvas pixels and red/green tests; unsuccessful optimizations remain visible. Humans supplied requirements, visual feedback and deployment access; another agent completed Dashboard. Human NVDA/VoiceOver review and changed-deadline/live-discussion cancellation were not verified. README, docs/performance-results.md and reports/evidence-manifest.json provide reproduction and scope. The user chose to send the local submission email personally.